

Software Requirements Specification (SRS)

Smart Learning Platform for Kids (Class 1 to 5) with Parent Progress Monitoring

1. Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) is to define the functional and non-functional requirements for the **Smart Learning Platform for Kids (Class 1 to 5) with Parent Progress Monitoring**.

The platform is designed to provide a safe, interactive, and engaging digital learning environment for primary school students. It helps children learn through animated lessons, videos, quizzes, and games while enabling parents to monitor their child's academic progress through reports and notifications.

The application supports only **four subjects**:

- English
 - Mathematics
 - Environmental Studies (EVS)
 - Basic Computer Knowledge
-

1.2 Project Overview

The Smart Learning Platform is a web and mobile application developed for **students from Class 1 to Class 5 (Age: 6–11 years)**.

The platform allows students to:

- Learn through interactive lessons

- Watch educational videos
- Solve quizzes

Parents can monitor:

- Daily learning activities
- Quiz scores
- Weekly and monthly reports

The Administrator manages students, parents, educational content, quizzes, and reports.

1.3 Scope

The Smart Learning Platform provides digital education for primary school students with parent monitoring features.

The system includes:

- Student learning dashboard
- Parent monitoring dashboard
- Administrator dashboard
- Subject-wise quizzes
- Progress tracking
- Notifications
- Performance reports

Supported Subjects

1. English
 2. Mathematics
 3. Environmental Studies (EVS)
 4. Basic Computer Knowledge
-

1.4 Objectives

The objectives of the system are:

- Make learning interesting and enjoyable.
- Improve children's basic knowledge.
- Encourage daily learning habits.

- Help parents monitor learning progress.
 - Automatically generate performance reports.
 - Provide a secure and child-friendly learning environment.
-

1.5 Intended Users

- Student (Class 1–5)
 - Parent
 - Administrator
-

1.6 Definitions

Term	Description
Student	Child using the application
Parent	Guardian monitoring child's progress
Administrator	Person managing the application
Dashboard	Main screen showing reports and progress
Quiz	Short assessment after every lesson

2. Overall Description

2.1 Product Perspective

The Smart Learning Platform is a cloud-based educational system.

It consists of three modules:

- Student Module
- Parent Module
- Admin Module

The application works on:

- Android
 - Windows
 - Web Browser
-

2.2 Product Features

Student Module

- Student Registration
 - Login
 - Profile Management
 - Class Selection (Class 1–5)
 - Subject Selection
 - Animated Lessons
 - Educational Videos
 - Practice Quizzes
 - Progress Tracking
-

Parent Module

- Parent Registration
 - Login
 - Link Child Account
 - View Child Dashboard
 - Subject-wise Progress
 - Quiz Results
 - Learning Time
 - Daily Report
 - Weekly Report
 - Monthly Report
 - Notifications
-

Admin Module

- Login
- Manage Students

- Manage Parents
 - Manage Subjects
 - Upload Videos
 - Upload Study Material
 - Create Quizzes
 - Manage Reports
 - Send Notifications
 - Backup Database
-

2.3 User Classes

User	Responsibilities
Student	Learn lessons and complete quizzes
Parent	Monitor child's progress
Administrator	Manage complete platform

2.4 Operating Environment

Hardware

- Smartphone
- Tablet
- Laptop
- Desktop Computer

Software

- Android
 - iOS
 - Windows
 - Chrome
 - Edge
 - Firefox
-

2.5 Constraints

- Internet connection required
 - Secure authentication
 - Child-friendly interface
 - Mobile responsive
 - Fast loading
-

3. Functional Requirements

FR1 Student Registration

Students register using:

- Name
- Age
- Class
- Parent Mobile Number
- Parent Email
- Password

Output:

Student account created successfully.

FR2 Parent Registration

Parents register using:

- Name
- Mobile Number
- Email
- Password

Output:

Parent account created successfully.

FR3 Login

Students, Parents, and Admin can login securely using:

- Email
 - Password
-

FR4 Student Dashboard

The dashboard displays:

- Available Subjects
 - Completed Lessons
 - Quiz Scores
-
- Learning Progress
 - Daily Goal
-

4. Non-Functional Requirements

Security

- Password Encryption
 - Secure Login
 - Role-Based Access Control
 - HTTPS Communication
 - Regular Database Backup
-

Reliability

- 99% availability

- Automatic backup
 - Error recovery support
-

Usability

- Child-friendly interface
 - Colorful design
 - Easy navigation
 - Large icons and buttons
-

Maintainability

- Modular architecture
 - Easy bug fixing
 - Easy software updates
-

Scalability

Future support for:

- AI Learning Assistant
 - Offline Learning
 - Regional Languages
-

7. Use Case Diagram (Actors)

Student

- Register
- Login
- Select Class
- Learn Subjects
- Watch Videos
- Attempt Quiz

- View Progress
-

Parent

- Register
 - Login
 - Link Child Account
 - Monitor Progress
 - View Reports
 - Receive Notifications
-

Administrator

- Login
 - Manage Students
 - Manage Parents
 - Manage Subjects
 - Upload Lessons
 - Manage Quizzes
 - Generate Reports
-

9. Future Enhancements

- AI-based personalized learning recommendations.
 - Voice-assisted reading for English lessons.
 - Interactive mathematics games with adaptive difficulty.
 - Virtual EVS activities and simulations.
 - Offline learning mode.
 - Digital certificates and achievement rewards.
 - Regional language support.
 - AI chatbot for answering basic student questions.
 - Enhanced analytics for parents to identify strengths and areas for improvement.
-

10. Assumptions and Dependencies

- Students have access to a smartphone, tablet, or computer with an internet connection.
 - Parents create and link their account with their child's account.
 - Learning content is uploaded and maintained by the administrator.
 - The system uses a secure database to store user information and learning records.
 - Users have basic knowledge of operating a mobile app or web browser.
-

11. Conclusion

The **Smart Learning Platform for Kids (Class 1 to 5) with Parent Progress Monitoring** is designed to provide an engaging, interactive, and secure educational environment for young learners. By focusing on **English, Mathematics, Environmental Studies (EVS), and Basic Computer Knowledge**, the platform strengthens foundational learning through animated lessons, quizzes, educational games, and rewards. Parents can actively monitor their child's progress with real-time reports and notifications, while the administrator efficiently manages content and users. The platform aims to improve learning outcomes, encourage consistent study habits, and build a strong foundation for future education.